AI for Sustainable Leadership — Frameworks Cheat Sheet

Exam-ready. 4 content blocks (S2–S17), ~40 frameworks, ~20 visual diagrams. Cite by name + author + the move it gives you. Northern light: mock exam (EU AI Act · RACI · resistance to change · Hahn et al. · sourcing).

How to use

- 🔴 **Core** ★★★★★ — essay anchors. Every long answer should rest on at least one of these.
- 🔵 **Key** ★★★★ — strong supporting cites. Use to add nuance or a second voice.
- 🟣 **Supporting** ★★★ — exam-detail level. Drop in to show range.
- One-line italic under the name = the **one-breath argument** to lead with.
- Bullets = the **moves** (what to actually write).

Block 1 · Framing AI as a Leadership & Sustainability Challenge (S2–S3)

🔴 Integrative Sustainability Framework · S2 · Hahn et al. (2014) ★★★★★

Sustainability is the simultaneous management of economic, environmental and social tensions across three levels and across time and space.

3 dimensions × 3 levels (+ temporal + spatial dependencies)

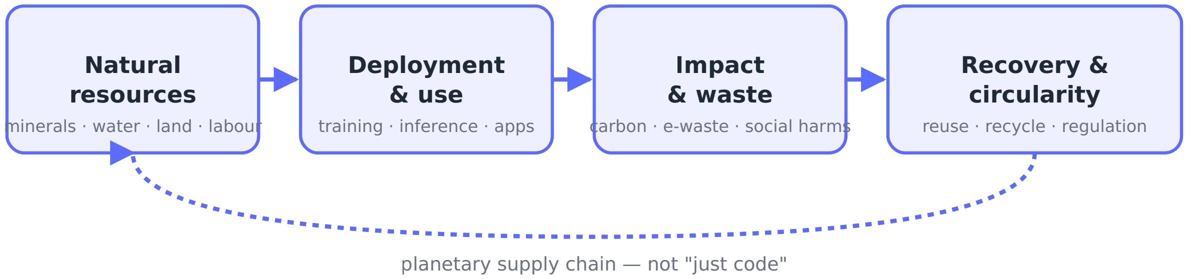
Level	Economic	Environmental	Social
Individual			
Organizational			
Systemic			

□ Temporal dependency — past inequalities encoded into present + projected forward
□ Spatial dependency — org embedded in & co-produces the wider system

- **3 dimensions** — economic / environmental / social (must hold together, no trade-offs assumed away)
- **3 levels** — individual / organizational / systemic (the same tension shows up in all three at once)
- **Temporal dependency** — present actions encode past inequalities and shape future capability
- **Spatial dependency** — orgs are embedded in and co-produce the wider system; "we're neutral" misrepresents the role
- **Exam move (mock Q2.1)** — identify the tension, name all three levels, then name temporal + spatial dependencies before proposing a response

AI Ecosystem Map · S2 · Crawford (2021) · Gambacorta & Shreeti (2025) ★★★★★

AI is not "just code" — it is a planetary supply chain: natural resources → deployment & use → impact & waste → recovery & circularity.



- **Natural resources** — minerals, water, energy, land, human labor (data labeling, ghost work)
- **Deployment & use** — model training, inference, applications, infrastructure
- **Impact & waste** — carbon emissions, e-waste, water stress, social externalities
- **Recovery & circularity** — reuse, recycling, regulation, redistribution
- **Exam move** — when asked about AI sustainability, name the full ecosystem instead of only emissions

Sustainability Challenges in AI · S2 · Crawford / Gambacorta & Shreeti ★★★★★

Four families of harm produced by the AI ecosystem.

- **Environmental** — carbon, water stress, resource depletion, e-waste, environmental injustice
- **Economic & geopolitical** — vendor lock-in, geopolitical risk, hidden liabilities, regulatory risk
- **Labor & inequality** — skill polarization, unequal economic gains, innovation inequality, reduced diversity, de-skilling, ghost work, crowdworking, job reduction, task erosion
- **Power & rights** — surveillance, consent fatigue, power asymmetries, biased outcomes, social exclusion, loss of trust
- **Exam move** — list 2 from each family to show coverage, not 5 from one

Sustainability Barriers in AI · S2 ★★★★★

Why even well-intentioned actors fail to act sustainably.

- **Market structure** — extreme concentration, high entry barriers, fixed costs, unequal bargaining power
- **Incentives** — misaligned (social/env secondary), short-terminism, investor pressure
- **Regulation** — fragmented across jurisdictions, regulatory lag
- **Supply chain** — geographic fragmentation, diffused responsibility
- **Organization** — functional silos, no single owner
- **Exam move** — use "barriers" to explain why sustainable leadership requires *system change*, not just intent

● Role of Sustainable Leaders · S2 ★★★★★

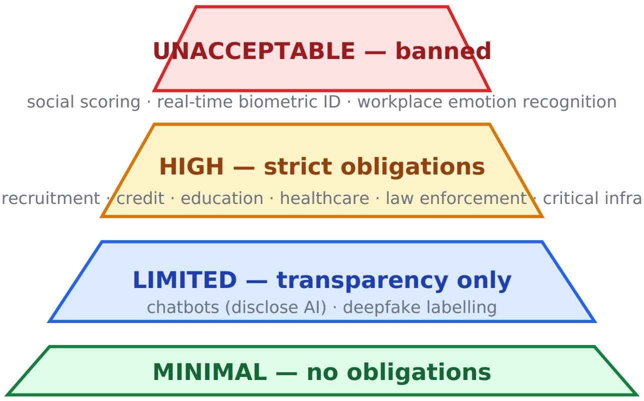
Sustainable leaders manage tensions across time and stakeholders rather than maximize one dimension.

- **Reduce tensions in the long term** — design out conflicts before they recur
- **Decrease number & magnitude of trade-offs** — engineer win-wins where possible
- **Align interests** — across social, environmental, economic concerns at different levels
- **"Exceed narrowly, meet broadly"** — pick a focus where you can lead; meet baseline elsewhere
- **Exam move** — use this in any "what should the leader do?" prompt as the lens

● EU AI Act — Risk Tiers · S3 ★★★★★

The EU classifies AI systems by risk level; obligations scale with risk.

Higher risk = stricter obligations



- **Unacceptable risk** — banned outright (social scoring by governments, real-time biometric ID in public, manipulative AI, emotion recognition in workplace/education)
- **High risk** — strict obligations: conformity assessment, risk management, human oversight, logging, transparency (employment/recruitment, education, credit scoring, law enforcement, critical infrastructure, biometrics)
- **Limited risk** — transparency obligations only (chatbots disclose AI, deepfake labeling)
- **Minimal risk** — no obligations beyond voluntary codes (spam filters, AI in video games)
- **Exam move (mock Q1.1)** — recruitment/HR AI = HIGH risk. The trap: human-in-the-loop does NOT downgrade the tier

● OECD AI Principles · S3 ★★★★

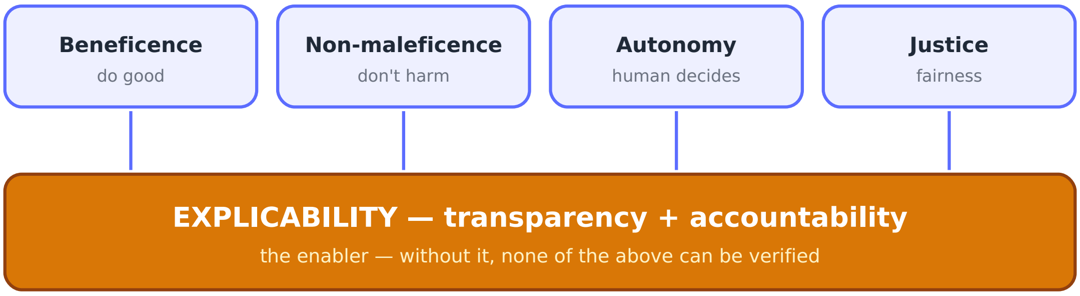
Five non-binding principles for trustworthy AI, adopted internationally.

- **Inclusive growth, sustainable development & well-being** — AI for people and planet
- **Human-centred values & fairness** — respect rule of law, human rights, diversity, equity
- **Transparency & explainability** — disclose AI use, make outputs understandable
- **Robustness, security & safety** — sound throughout lifecycle, traceable, manageable risk
- **Accountability** — organizations responsible for functioning + adherence to the above
- **Exam move** — cite OECD when the question asks about international/normative frameworks

Floridi & Cowls (2019) — Unified Ethics Framework ★★★★

Five principles that synthesize 47 published AI ethics statements.

4 classical bioethics principles + 1 AI-specific enabler



- **Beneficence** — promote well-being, preserve dignity, sustain planet
- **Non-maleficence** — privacy, security, capability caution
- **Autonomy** — human power to decide, to delegate, and to revoke delegation
- **Justice** — promote prosperity, preserve solidarity, avoid unfair discrimination
- **Explicability** — enables the other four (transparency + accountability)
- **Exam move** — the "5th principle is the enabler" framing — explicability ≠ a goal, it's the precondition

Ethical AI Landscape — 4 Framework Families · S3 ★★★★

The full toolkit of AI ethics responses; questions often ask which family applies.

- **Regulatory & legal** — EU AI Act (hard law)
- **Normative & principles-based** — OECD, UNESCO, company AI principles (soft norms)
- **Process & lifecycle** — responsible AI lifecycle, ethics-by-design, model cards, data sheets
- **Governance & accountability** — AI ethics committees, AIA / DPIA (algorithmic / data protection impact assessments), Human-in-the-loop (HITL), Human-on-the-loop (HOTL), red-teaming, audits
- **Exam move** — when asked "what would you do?" list responses from at least 2 families

Transformational Leadership · S3 · Bass & Riggio (2006) ★★★★

Leaders inspire followers through vision and values rather than transactional reward.

- **4 I's** — Idealized influence, Inspirational motivation, Intellectual stimulation, Individualized consideration
- **Why it matters for AI** — sets a vision that integrates ethics into the why, not just the what
- **Exam move** — use as the "vision-setting" anchor in change scenarios

Servant Leadership · S3 · Dierendonck & Nuijten (2011) ★★★★

Leader prioritizes follower development and stakeholder welfare over self-interest.

- **Empowerment, stewardship, humility, accountability, courage, forgiveness, authenticity, standing back**
- **Why it matters for AI** — counters power-asymmetry risk of AI by centering the people affected
- **Exam move** — pair with sustainability questions about employee voice and stakeholder care

Ethical Leadership · S3 · Brown et al. (2005) ★★★★

Demonstration of normatively appropriate conduct through personal actions and interpersonal relationships, plus promotion of such conduct to followers.

- **Moral person** — traits + behavior the leader embodies
- **Moral manager** — communicates ethics, rewards/disciplines, sets the climate
- **Why it matters for AI** — "tone at the top" decides whether ethics policies become real
- **Exam move** — cite when the question is about *climate* or *culture* of responsible AI

Moral Courage · S3 · Edmondson (2020) · Hannah & Avolio (2010) ★★★★

The capacity to act on ethical convictions despite risk of loss.

- **Required for "speaking up"** — in psychologically unsafe AI deployment contexts
- **Connects to psychological safety** — courage is needed where safety is absent
- **Exam move** — use to explain why ethics policies fail without leadership courage

Wade & Tomoko (2024) · Blackman (2022) — Building AI Ethics Capability · S3 ★★★

Ethics has to be an organizational capability, not a checklist.

- **Translate principles into practical guidance** — turn "fairness" into testable criteria
- **Integrate into AI design and development** — embed in lifecycle, not append at end
- **Adapt to local conditions** — context-specific implementation
- **Share best practices across organization** — institutional learning loop
- **Multidisciplinary AI ethics committee** — tech + business + legal + affected groups
- **Exam move** — cite when answering "how do we move from principles to practice?"

Czakert & Wietrak (2025) — New Leadership for AI · S3 ★★★

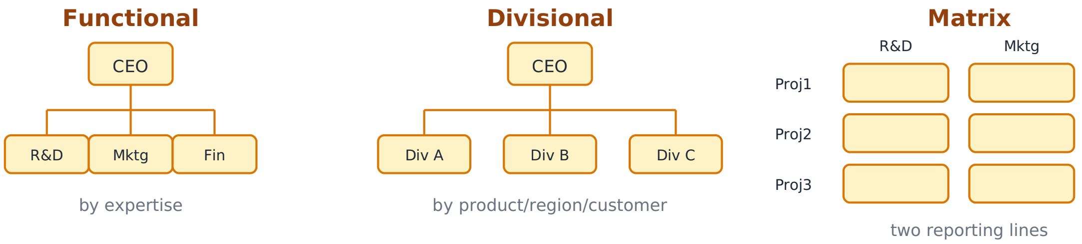
Existing leadership theories treat ethics as a supplemental virtue; AI context needs ethics as foundational.

- **Old framing** — ethics added on top of governance
- **New framing** — ethics designed into the system; leaders accountable at systemic level
- **Exam move** — cite as the "we need a new model" voice when contrasting traditional leadership theories

Block 2 · Designing AI Organizations (S4–S7)

Organizational Structure — Basic Forms · S4 ★★★★★

The frame of reporting relationships that divides + coordinates work.



Differentiation (divide work) + Integration (coordinate) are the two purposes of any structure

Functional = deep specialization, weak cross-coord · Divisional = autonomy, duplication · Matrix = flexible, conflict-prone

- **Functional** — grouped by expertise (R&D, marketing); efficient, deep specialization, weak cross-coord
- **Divisional** — grouped by product/region/customer; high autonomy, duplication of resources
- **Matrix** — two reporting lines (e.g., function + project); flexible, prone to conflict and slow decisions
- **Differentiation & Integration** — structure must both divide work and coordinate it
- **Exam move** — name the form, then say what it OPTIMIZES (control / speed / specialization)

Vertical (Embedded) vs Horizontal (Centralized) AI Teams · S4 ★★★★★

Two patterns for placing AI capability inside a company; each has a signature trade-off.

3 tensions

Speed vs Consistency · Local Relevance vs Global Standards · Business Ownership vs Technical Ownership

- **Vertical / Embedded** — AI lives inside business units; high contextual relevance, weak consistency
- **Horizontal / Centralized** — AI sits in a central function; standards + scale, but distance from business
- **Three tensions** — Speed vs Consistency · Local Relevance vs Global Standards · Business Ownership vs Technical Ownership
- **Exam move** — diagnose the org's *pain* (slow rollouts? inconsistency? irrelevant pilots?) → prescribe the form

Alternative Structures — Amorphous & Holacracy · S4 ★★★★

Modern responses to the rigidity of functional/divisional/matrix.

- **Amorphous** — fluid, unstructured; teams form/dissolve around problems
- **Holacracy** — structured self-management; rules + roles replace managers
- **Pros** — autonomy, faster decisions, adaptability, transparency, reduced bureaucracy
- **Cons** — complex to implement, steep learning curve, role confusion, doesn't suit regulated industries
- **Exam move** — useful when question asks for "alternative to hierarchy"

Organizing Paradox & Job Deconstruction · S4 ★★★★

Balance the need for structure with the need for flexibility by matching skills to tasks dynamically.

- **Job deconstruction** — replace fixed roles with skill-task matching
- **Skills-first organizations** — Delta Airlines example; recruit on skills not credentials
- **Open Opportunities** (U.S. federal gov) — internal marketplace for skills
- **Exam move** — cite when question is about future of work or AI augmentation

AI Lab Archetypes · S5 ★★★★★

Three structural patterns for AI labs; each encodes trade-offs between speed, coherence, accountability and time horizon.

CENTRALIZED	DECENTRALIZED	VENTURE / INCUBATOR
"pool the talent"	embedded in units	"startup logic"
Reports to senior leadership	AI inside business units	Speed, autonomy, external
Standardized practice	High contextual relevance	Built for exploration
✓ strong control	✓ relevance	✓ exploration
✗ distance from biz	✗ coordination gaps	✗ not for incremental

No archetype is superior — each encodes trade-offs between speed, coherence, accountability, time horizon

- **Centralized Lab** — pools scarce talent, standardizes practice, reports to senior leadership. Strong control, risks distance from business
- **Decentralized / Embedded Lab** — AI inside business units. High contextual relevance, weak coordination + consistency
- **Venture / Incubator Lab** — start-up logic: speed, autonomy, external orientation. Built for exploration, not incremental improvement
- **No archetype is superior** — each is a trade-off
- **Exam move** — diagnose the org's strategic intent first (explore vs exploit) → choose archetype

Hybrid AI Labs & Social Alignment · S5 ★★★★★

Hybrid labs explicitly pursue both business and social objectives — alignment is structural, not cosmetic.

- **Why they emerge** — external pressure: regulation, scrutiny, ethics, legitimacy
- **What alignment means** — embed social/env considerations into HOW problems are selected, HOW success is measured, HOW trade-offs are resolved — *from day one*
- **Exam move** — alignment ≠ a CSR layer; if it's added after the fact, it isn't alignment

Innovation Theatre · S5 ★★★★★

Organizations adopt the symbols of innovation without changing the underlying power, incentives, or decision rights.

- **The symbols** — impressive lab spaces, demos, mission statements, polished prototypes, bold AI strategies
- **What doesn't change** — decision rights stay concentrated, incentives still reward short-term exploitation, pilots never reach production
- **The risk** — erodes credibility, misallocates talent, delays genuine capability building
- **Not deliberate deception** — emerges from unresolved tensions (explore/exploit, signaling/operational, long-term/short-term)
- **Exam move** — name when an "AI initiative" is mostly PR; pair with hybrid alignment to show real vs theatre

Make / Buy / Partner — Sourcing AI Capabilities · S6 ★★★★★

Three paths to acquire AI capability, each with a signature risk profile.

MAKE (build)	BUY (license)	PARTNER (co-dev)
✓ ADVANTAGES	✓ ADVANTAGES	✓ ADVANTAGES
• Control + differentiation	• Quick to deploy	• Shared experimentation
• Long-term learning	• Lower upfront cost	• Complementary skills
• Protects strategic assets	• Focus on core	• Cross-domain learning
• Value-aligned governance	• Scalability	• Flexibility
✗ RISKS	✗ RISKS	✗ RISKS
• Slow + expensive	• Capability erosion	• Complex to manage
• Talent constraints	• Vendor dependency	• Power asymmetries
• Tech obsolescence	• Switching costs high	• IP / data lock-in
• Overconfidence	• Blurred accountability	• Different priorities
→ strategic differentiator	→ non-core, need speed	→ complementary expertise

- **MAKE (build)** — control, contextual learning, protects assets, differentiation possible, adaptive governance. BUT: slow, expensive, talent constraints, risky (obsolescence)
- **BUY (license vendor)** — quick, lower upfront, risk shifted, focus on core. BUT: capability erosion, opacity, dependency, switching costs, misaligned vendor incentives, blurred accountability, regulatory exposure, ethical delegation without control
- **PARTNER (co-develop)** — shared experimentation, complementary capabilities, cross-domain learning, flexibility. BUT: complex management, power asymmetries, ambiguous governance, IP/data lock-in, different priorities
- **Exam move (mock Q3.1)** — diagnose: core/non-core? time? talent? values? Then pick — non-core + speed + no infra = BUY; strategic differentiator = MAKE; complementary expertise needed = PARTNER

● AI Innovation Strategies — Four Paths · S6/S7 · Bouquet et al. (2026) ★★★★

Strategic typology of AI innovation across organizations.

- **Four paths** mapping the build/buy/partner spectrum to strategic intent
- **Use** — assess where an organization sits and where it should move
- **Exam move** — pair with Make/Buy/Partner to give a *strategic* framing on top of the *transactional* one

● Portfolio Management · S7 · Kloppenborg & Lanning (2012) ★★★★★

Centralized management of multiple projects to meet strategic business objectives.

5 executive tasks · sequential

Identify (scan + name) → **Select** (choose set) → **Prioritise** (rank order) → **Resource** (budget + talent) → **Govern** (control loop)

Strategy development → Portfolio management → Project management

- **Portfolio** = collection of projects that can be measured, ranked, prioritized
- **Portfolio mgmt** = identifying, prioritizing, authorizing, managing, controlling projects to achieve strategy
- **Executive tasks (5)** — Identify · Select · Prioritize · Resource · Govern
- **Hierarchy** — Strategy development → Portfolio mgmt → Project mgmt
- **Exam move** — distinguish PROJECT mgmt (delivering one) from PORTFOLIO mgmt (allocating across many)

● Portfolio Stress-Testing — Scenario Lenses · S7 (NordSteel case) ★★★★

Four pressure questions to expose hidden fragility in an AI portfolio.

- **Cost shock** — if API/compute costs rise 25%, does the portfolio still work?
- **Regulatory shock** — if new explainability rules hit, are you exposed?
- **Competitive shock** — if a competitor launches a proprietary foundation model, what happens?
- **Identity shock** — what type of company does your portfolio imply you'll become in 5 years?
- **Exam move** — use as a checklist to evaluate any portfolio scenario in the exam

Block 3 · People & Leadership in AI Organizations (S8–S12)

Task / Role / Job / Authority / Profile · S8 ★★★★★

Five distinct concepts confused at most companies; clarity here drives accountability.

Task	Role	Job	Authority	Profile
what is done (action)	responsibility in the system	formal position (HR title)	decision rights	who can do it well (skills)

"AI Product Owner responsible for model success" → which of the 5 is unclear?

- **Task** — what is done (concrete action)
- **Role** — responsibility in the system (what the person owns)
- **Job** — formal position in the organization (the HR designation)
- **Authority** — decision rights (what the person can decide alone)
- **Profile** — who can perform the role successfully (skills + personality)
- **Exam move** — when a job description is unclear, name which of these is missing

AI Team Role Map · S8 ★★★★

Four families of roles in AI organizations.

- **Core Technical** — Data Scientist, ML Engineer, MLOps Engineer, Data Engineer, Technical Lead / AI Architect
- **Product & Integration** — AI Product Owner / AI PM, AI Translator, Domain Expert (SME), UX / Human-Centered AI Designer
- **Governance & Risk** — AI Governance Lead, Compliance/Legal, Responsible AI/Ethics Officer, Model Risk Manager, Data Protection Officer
- **Strategic & Leadership** — Chief AI Officer / Head of AI, CTO, Business Sponsor, Chief Data & AI Officer (CDAIO)
- **Exam move** — diagnose missing roles when team gaps show up

Recruitment vs Selection · S9 ★★★★

Two complementary HR processes that must align with org strategy.

- **Recruitment** — attracting and identifying candidates; goal: create awareness + attract diverse range
- **Selection** — evaluating and choosing candidates; goal: hire the best fit (job + culture)
- **Key principle** — alignment of both with organizational goals AND employee expectations
- **Exam move** — they sound similar; the distinction is *attract* vs *choose*

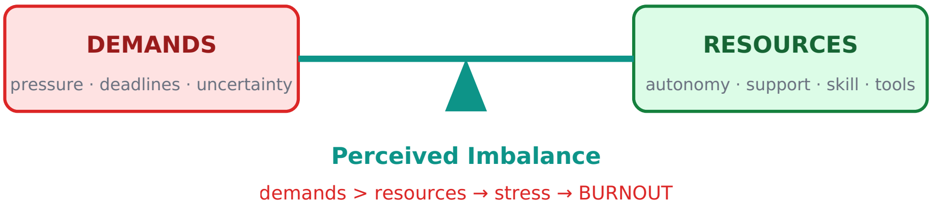
Quality of Hire & RippleHire · S9 ★★★

AI-driven recruitment platform showing what ethical AI in HR looks like.

- **Quality of hire** = measurable post-hire performance + retention + cultural fit
- **Ethical AI in recruitment** — explainability, bias auditing, candidate experience
- **Exam move** — cite RippleHire as a positive case (vs Q1.1 hiring AI as a high-risk one)

Job Demands–Resources Model · S10 · Bakker & Demerouti (2006/2007) ★★★★★

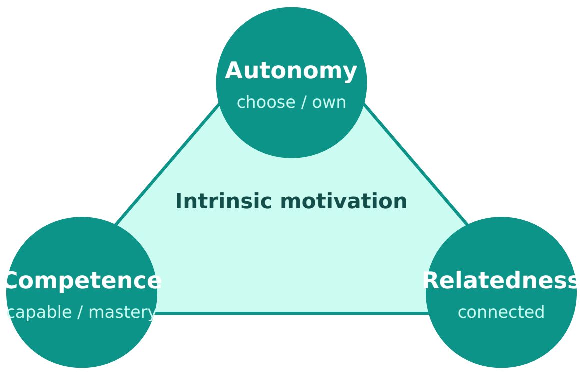
Stress and burnout emerge from a perceived imbalance between demands and resources.



- **Demands** — responsibilities, pressures, obligations, uncertainties at work
- **Resources** — things within the individual's control to resolve demands
- **Imbalance** — chronic demand > resource → stress → burnout
- **Connected research** — Karasek & Johnson (1986), Le Blanc et al (2008)
- **Exam move** — diagnose AI-workplace burnout (e.g., AI overload) with this model

Self-Determination Theory (SDT) · S10 · Deci & Ryan ★★★★★

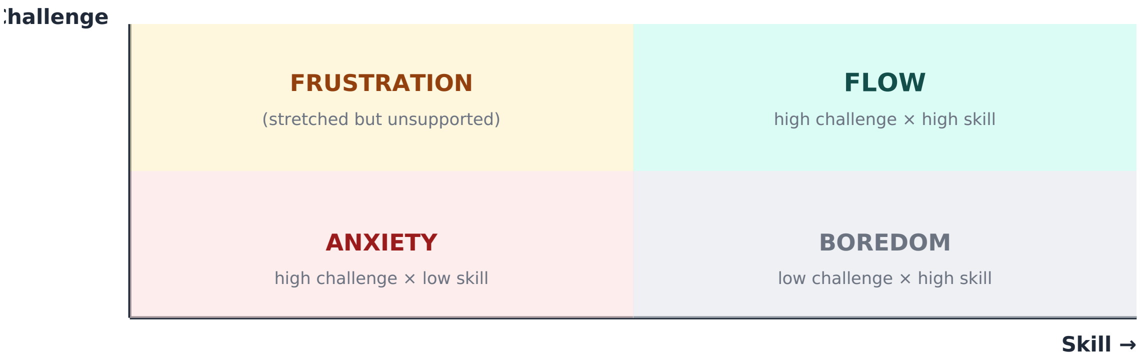
Intrinsic motivation requires three psychological needs to be met.



- **Autonomy** — feeling of volition over actions
- **Competence** — feeling effective in challenges
- **Relatedness** — feeling connected to others
- **Exam move** — diagnose why AI tools demotivate workers (e.g., undermining autonomy/competence)

Flow Theory · S10 · Csikszentmihalyi ★★★★

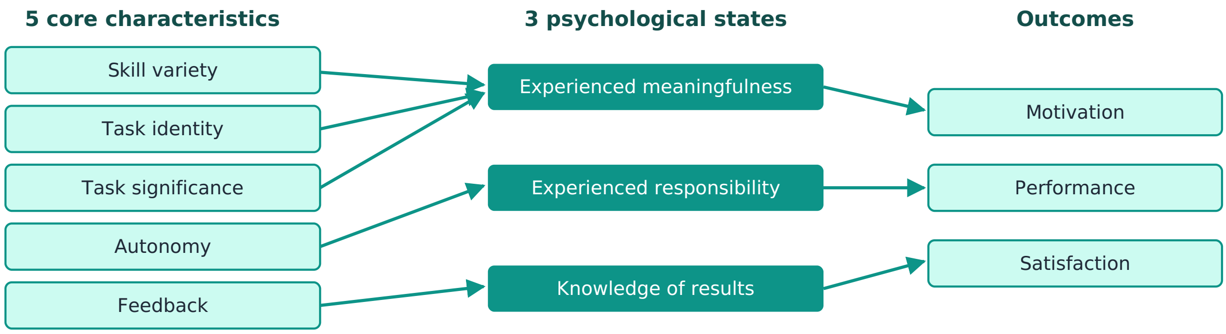
Optimal experience occurs when challenge matches skill.



- **Flow zone** — high challenge × high skill
- **Boredom** — low challenge × high skill
- **Anxiety** — high challenge × low skill
- **Exam move** — use for AI job redesign questions (where should AI take routine tasks?)

Job Characteristics Model · S10 · Hackman & Oldham ★★★★★

Five core characteristics drive motivation and satisfaction.



Use to redesign AI-affected jobs: where can AI absorb monotony without killing autonomy?

- **Skill variety** — range of skills required
- **Task identity** — visible whole task
- **Task significance** — impact on others
- **Autonomy** — freedom in how
- **Feedback** — direct info on performance
- → produce **experienced meaningfulness, responsibility, knowledge of results** → motivation, performance, satisfaction
- **Exam move** — use when asked how to redesign a job AI is reshaping

● Psychological Contract · S10 · Rousseau (1995) ★★★★★

Individual's belief about the terms of a reciprocal exchange with the employer.

- **Promise** — explicit or implicit, both parties
- **Consideration offered** — what each gives in return
- **Reciprocal obligations** — bind the parties
- **Why it matters with AI** — AI changes job content; the implicit promise may break ("I was hired to think, now I supervise the model")
- **Exam move** — use when an AI rollout causes morale collapse without obvious explanation

● Psychological Safety · S10 · Edmondson (1999) ★★★★★

Shared belief that the team is safe for interpersonal risk-taking.

- **Required for** — speaking up about errors, asking questions, suggesting changes, taking risks
- **In AI contexts** — critical for flagging AI errors, model bias, deployment risks
- **Exam move** — link to moral courage (Edmondson 2020); without safety, courage is solitary

● People Analytics — Definition · S10 · Ferrar & Green (2021) ★★★★★

The analysis of employee and workforce data to reveal insights and provide recommendations to improve business outcomes.

- **Applies to** — recruitment, performance, engagement, retention
- **Also called** — HR analytics, workforce analytics
- **Key shift** — from intuition-driven to data-informed decision-making
- **Exam move** — open with this definition any people-analytics essay; it's the standard reference

● Analytics Activities → Outcomes (Ferrar & Green) · S10 ★★★★

Four levels of analytics activity producing four families of outcomes.

Activities (maturity ↑)		Outcomes
1 · Dashboards & reporting		Workforce experience
2 · KPIs & metrics	→	Analytics culture
3 · Predictive analytics		Business performance
4 · Advanced analytics & AI		Societal benefits

Dashboards alone ≠ analytics · Advanced AI without analytics culture = wasted spend

- **Activities** — Dashboards & reporting · KPIs & metrics · Predictive analytics · Advanced analytics & AI
- **Outcomes** — Workforce experience · Analytics culture · Business performance · Societal benefits
- **Exam move** — show progression: dashboards alone ≠ analytics; advanced AI without culture = waste

● Reporting vs Analytics · S11 ★★★★★

The key distinction that defines People Analytics as a discipline.

- **Reporting** — tells you turnover was 15%
- **Analytics** — tells you turnover was 15% BECAUSE new sales hires lack effective onboarding — and here's how to fix it
- **Reporting answers "what"; analytics answers "why" and "what next"**
- **Exam move** — use this as your one-line definition under exam pressure

● Organizational Network Analysis (ONA) · S11 ★★★★

Maps actual collaboration patterns (who talks to whom) vs the formal org chart.

- **Reveals** — bottlenecks, hidden influencers, silos, brokers
- **Inputs** — emails, calendars, surveys, communication metadata
- **Risks** — surveillance, privacy, manipulation
- **Exam move** — pair with ethical risk discussion (consent, transparency, purpose limitation)

● People Analytics as Capability · S11 · Adam Grant quote ★★★

"Every opinion you hold at work is a hypothesis waiting to be tested."

- **Move** — from analytics as a specialized FUNCTION to analytics as an organizational CAPABILITY
- **Exam move** — use to argue against siloed "analytics teams"

● Kahneman — System 1 / System 2 · S12 · Kahneman (2002 Nobel) ★★★★★

Two modes of thinking that explain systematic biases in human judgment.

SYSTEM 1 — fast

"thinking without thinking"

- automatic
- emotional · intuitive
- effortless
- face recognition
- loud-noise reaction

SYSTEM 2 — slow

"thinking, hard"

- deliberative
- logical · effortful
- requires attention
- hard maths
- narrow parking

- **System 1 (Fast)** — automatic, emotional, intuitive, effortless (face recognition, loud noise reaction)
- **System 2 (Slow)** — deliberative, logical, effortful (hard math, narrow parking)
- **Implication** — humans are NOT the rational decision-makers traditional economics assumed
- **AI implication** — AI substitutes for System 1 speed; the danger is humans skip System 2 oversight
- **Exam move** — cite when asked about AI augmentation, bias, or decision quality

Cognitive Surrender ("Confidently Wrong") · S12 ★★★★★

When AI is accurate, humans improve a lot; when AI is wrong, humans drop hard — they trust the model even when it misleads.

- **Mechanism** — over-reliance on AI; human oversight collapses into rubber-stamping
- **Risk** — automated decision becomes worse than either human or AI alone
- **Mitigation** — keep humans in deliberative System 2 role; design for *constructive friction*
- **Exam move** — use to argue against passive HITL designs; human oversight must be active

AI for Leadership — Benefits & Trade-offs · S12 ★★★

How AI augments leadership work, and where it risks displacing judgment.

- **Benefits** — better decision quality + speed, less routine, more time for strategy + people, AI coaching, tailored communication, frequent feedback, bias audits, workload monitoring
- **Trade-offs** — cognitive surrender, loss of judgment muscle, surveillance of teams, opaque criteria, accountability dilution
- **Exam move** — pair benefits with trade-offs; never one-sided

Block 4 · Executing, Scaling & Governing AI (S13–S17)

Digital Transformation Enablers vs Hindrances · S13 · Pajuoja (ManCo) ★★★★★

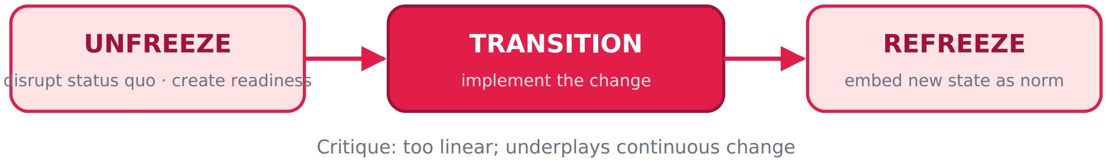
Four families of factors that enable or block GenAI adoption in non-AI-native environments.

Organization	Individual	Work context	GenAI tool
✓ **Enablers** • budget · structure • scaling capacity	✓ **Enablers** • positive attitude • willingness	✓ **Enablers** • competitive pressure • advantage-seeking	✓ **Enablers** • training available
✗ **Hindrances** • silos • regulated industry • cost/benefit gap	✗ **Hindrances** • lacks skills • reluctance to change • teaching effort	✗ **Hindrances** • expert knowledge • interdependencies • mistakes expensive	✗ **Hindrances** • tool restrictions • missing/bad data • lacks understanding

- **Organization** — budget, structure, scaling capability, structured rollout, cost-benefit; BUT functional silos, regulated industry constraints
- **Individual** — positive attitude, willingness; BUT lack of skills, reluctance to change, teaching effort
- **Work context** — competitive pressure, willingness to use for advantage; BUT importance of historical/expert knowledge, complex interdependencies, mistakes are expensive
- **GenAI tool** — training available; BUT tool restrictions, missing data, wrong format, lacks human understanding
- **Exam move** — diagnose adoption challenge by category, then propose actions per family

● Lewin's 3-Step Change Model · S14 · Lewin ★★★★★

The simplest, oldest change framework: unfreeze, change, refreeze.



- **Unfreeze** — disrupt the status quo, create readiness
- **Transition** — implement the change
- **Refreeze** — embed the new state as the norm
- **Exam move** — name when asked the "classic" change model; pair with critiques (too linear)

Kotter's 8-Step Change Model · S14 · Kotter ★★★★★

The action playbook for large-scale change.

Stouten et al. (2018) replace #1 with "diagnose the opportunity/problem" — urgency causes fear

1. **Establish a sense of urgency** *(weak scientific support — see Stouten critique)*
2. **Form a powerful guiding coalition**
3. **Create a vision**
4. **Communicate the vision**
5. **Empower others to act on the vision**
6. **Plan for and create short-term wins**
7. **Consolidate improvements and produce more change**
8. **Institutionalize new approaches**

- **Exam move** — name the 8 steps; flag that urgency creation is contested

ADKAR Model · S14 · Hiatt ★★★★★

Individual-level change model — change happens through individuals.

Individual-level lens (cf. Kotter's organizational lens)

- **A — Awareness** of the need to change
- **D — Desire** to participate and support
- **K — Knowledge** of how to change
- **A — Ability** to implement required skills/behaviors
- **R — Reinforcement** to sustain
- **Exam move** — cite as the *individual* lens vs Kotter's organizational lens

Stouten et al. (2018) — Evidence-Based 10 Steps · S14 ★★★★★

A synthesis of change research that critiques Kotter and proposes evidence-based steps. 1. **Assess the opportunity/problem** motivating change *(replaces "sense of urgency" — urgency causes fear/rejection)* 2. **Select and support a guiding change coalition** 3. **Formulate a clear compelling vision** 4. **Communicate the vision** *(43% less resistance if explained — Shaw et al. 2003)* 5. **Mobilize energy for change** *(Ability · Motivation · Opportunity — Ajzen 1991)* 6. **Empower others to act** *(procedural justice, voice — Oreg, Tyler & Blader)* 7. **Develop and promote change-related knowledge and ability** *(self-efficacy → motivation — Kao 2017)* 8. **Identify short-term wins** as reinforcements 9. **Monitor and strengthen the change process** *(feedback, course correction)* 10. **Institutionalize change** *(routines embedded — Edmondson; Rerup & Feldman)*

- **Exam move** — anchor change essays here; it's the evidence-based update to Kotter

Kotter & Schlesinger — 6 Approaches to Resistance to Change · S14 ★★★★★

Six tactics for overcoming resistance; choice depends on context, speed, and power.

Approach	When to use	Advantage	Drawback
Communication & education	Lack of (accurate) info	Once persuaded, people help implement	Time-consuming if many involved
Participation & involvement	Initiators lack info to design change	More commitment	Time-consuming if design is wrong
Facilitation & support	Adjustment problems (people fear they can't adjust)	Best for adjustment issues	Time-consuming, expensive, can still fail
Negotiation & agreement	Someone loses out and has power to resist	Relatively easy to avoid major resistance	Expensive if alerts others to negotiate
Manipulation & cooptation	Other tactics won't work / too expensive	Relatively quick + inexpensive	Future problems if people feel manipulated
Coercion	Speed is essential, initiators have power	Quick, overcomes any resistance	Resentment toward initiators

- **Exam move (mock Q1.3)** — match the approach to the constraint (e.g., senior leadership wants quick + tolerates some dissatisfaction → coercion or manipulation; if they want commitment → participation)

4 Foundations to Overcome Resistance · S14 ★★★★★

Beyond tactics, four organizational conditions that reduce resistance.

- **Develop positive relationships** — supervisor trust, change-agent trust, supportive environment → more positive feelings about change
- **Cultivate psychological safety** — safe interpersonal risk-taking → info sharing, learning, satisfaction
- **Implement changes fairly** — procedural justice; even if outcome is negative, impact is softened
- **Select people who accept change** — high openness, positive self-concept, risk tolerance, growth need, intrinsic motivation, internal locus of control
- **Importance of "local tribes"** — social systems matter for the change process
- **Exam move** — pair with Kotter & Schlesinger; tactics + foundations together

● Project Management Methodologies Map · S15 ★★★★★

Two families: plan-driven (predictive) vs adaptive (agile); pick by uncertainty + stability of requirements.

PREDICTIVE (low flex)		ADAPTIVE / AGILE (high flex)	
stable requirements · plan-driven		emergent requirements · learning-driven	
Waterfall	linear sequential · construction	**Scrum**	sprints + roles · cross-functional
CPM	critical path · large engineering	**Kanban**	visual flow · continuous delivery
PRINCE2	process-based · government	**Lean**	waste elimination · max value
Stage-Gate	stages with gates · NPD		

~85% of AI projects never reach production

"Primarily a management failure, not a tech failure" — Gartner

→ AI projects need ADAPTIVE methods, not classic predictive

- **Predictive (Plan-Driven, low flexibility)** —
  - **Waterfall** — linear, sequential; stable fixed requirements (construction)
  - **Critical Path Method (CPM)** — sequence of critical tasks; large engineering with dependencies
  - **PRINCE2** — process-based, structured stages, defined roles; government, large corporate
  - **Stage-Gate** — stages with quality/business gates; new product development
- **Adaptive (Agile umbrella, high flexibility)** —
  - **Scrum** — fixed sprints + defined roles; cross-functional teams, shifting work
  - **Kanban** — visual board, continuous flow; steady delivery without rigid cycles
  - **Lean** — eliminate waste, max value; process optimization
- **Exam move** — choose based on (a) uncertainty (b) requirement stability (c) what you want to optimize for

● Why AI Projects Fail (~85% never reach production) · S15 · Gartner ★★★★★

AI project failure is primarily a management failure, not a tech failure.

- **Data quality & availability** — #1 blocker (insufficient, biased, inaccessible)
- **Unclear business objectives** — no definition of success
- **Underestimated complexity** — assumed to work like traditional software
- **No plan for model drift** — silent degradation in production
- **Skill & culture gaps** — lack hybrid technical + business fluency
- **Exam move** — use the 85% stat as the hook; then list 2-3 reasons

● AI Project ≠ Classic Project · S16 ★★★★★

Five dimensions where classic PM assumptions break for AI.

Dimension	Classic project	AI project
Scope	Fixed & defined upfront	Emergent — shaped by data
Output	Concrete deliverable	Probabilistic model
Timeline	Predictable phases	Experiment-driven
Done when...	Features complete	"Good enough" accuracy
Failure mode	Visible bugs	Silent degradation in prod

Classic-PM assumptions fail on AI → need iterative discovery, not just execution

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Done when...	Features complete	"Good enough" accuracy
Failure mode	Visible bugs	Silent degradation in production

- **Exam move** — use this table when asked why classic PM fails on AI

● Good AI Project Leadership · S16 ★★★★★

Five practices that distinguish leaders who actually ship AI. 1. **Define success before you start** — concrete accuracy/ impact threshold 2. **Treat data as a deliverable** — not pre-project housekeeping 3. **Plan to fail fast** — explicit go/no-go gates and kill criteria 4. **Separate exploration from execution** — protect discovery from delivery pressure 5. **Make uncertainty visible to stakeholders** — communicate confidence, limits, risks

- **Exam move** — direct answer to "what makes AI project leadership good?"

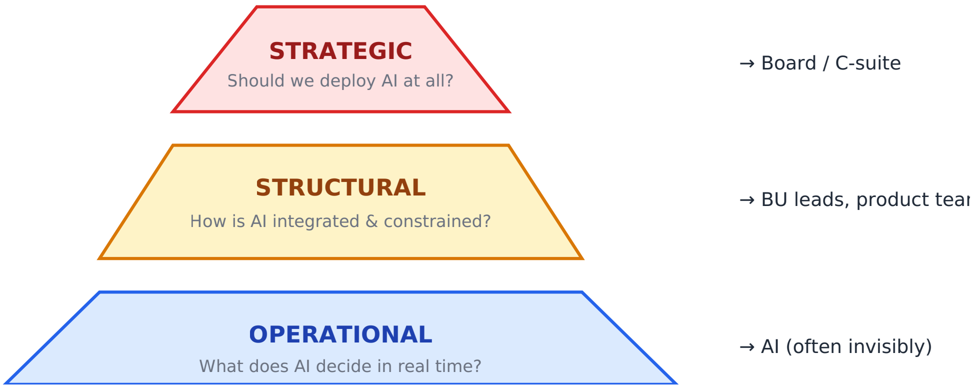
● 6Ps of Operational Capabilities · S16 ★★★

Six capability domains to assess when scaling AI.

- **People · Process · Products & Parts · Policy · Plant & Infrastructure · Partners**
- **Exam move** — use as a scan for capability gaps in scaling cases

● Decision Rights in AI — 3 Layers · S17 ★★★★★

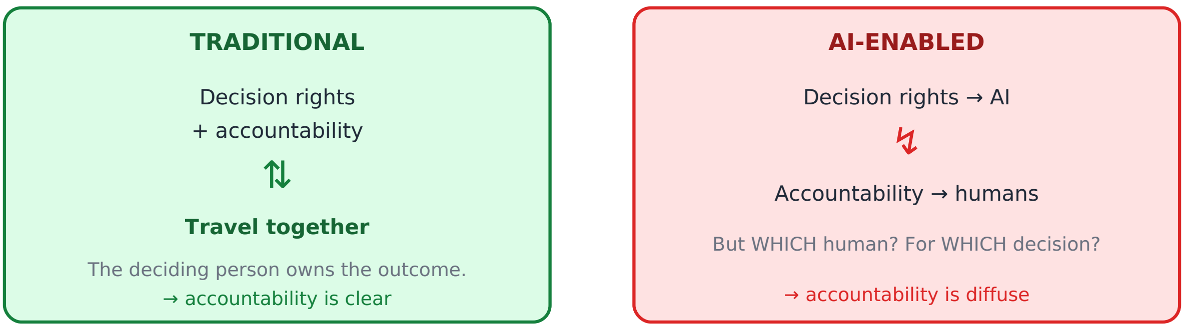
AI decisions sit at three levels; each has a different owner.



- **Strategic** — Should we deploy AI at all? → Board / C-suite
- **Structural** — How is AI integrated and constrained? → Business unit leads, product teams
- **Operational** — What does AI decide in real time? → Delegated (often invisibly) to the algorithm
- **Exam move** — pair with the decoupling problem to show where accountability evaporates

● The Decoupling Problem · S17 ★★★★★

Classic principle: decision rights + accountability travel together. AI breaks this.



- **Traditional** — person who decides owns the outcome → accountability is clear
- **AI-enabled** — decision rights delegated to AI; accountability stays with humans → but WHICH human? for WHICH decision?
- **Result** — accountability becomes diffuse
- **Exam move** — use as the *diagnostic frame* before naming a governance fix

Three Governance Archetypes · S17 ★★★★★

Three structural patterns for AI governance, each with a clear strength + weakness.

CENTRALIZED	FEDERATED	EMBEDDED
Central AI council / CAIO approves all deployments	Business units own AI within central policy	Governance baked into AI via guardrails + auto checks
✓ consistent standards ✓ clear accountability	✓ speed ✓ domain expertise + scale	✓ scales to volume ✓ consistent + fast
✗ bottleneck ✗ slow deployment	✗ policy drift ✗ accountability gaps	✗ who audits the guardrails?

- **Centralized** — central AI council/CAIO approves all deployments. STRENGTH: consistent standards, clear accountability. WEAKNESS: bottleneck — slow
- **Federated** — business units own AI decisions within central policy. STRENGTH: speed, domain expertise, scalable. WEAKNESS: policy drift, inconsistency, accountability gaps
- **Embedded** — governance baked into the AI system (guardrails, automated checks). STRENGTH: scales to volume, consistent, fast. WEAKNESS: who audits the guardrails?
- **Exam move** — pick by org's pain (speed → federated; rigor → centralized; volume → embedded)

RACI Framework · S17 ★★★★★

Maps accountability for any decision; four roles, only ONE is accountable.

R	A	C	I
Responsible	Accountable	Consulted	Informed
DOES the work	OWNS the outcome	input BEFORE decision	notified AFTER decision
can be multiple can be delegated	EXACTLY ONE role CANNOT delegate	two-way	one-way
(mock Q1.2: the DOER)	answerable when broken	no veto	not involved

- **R — Responsible** — does the work; CAN be multiple roles; responsibility CAN be delegated
- **A — Accountable** — owns the outcome; MUST be exactly ONE role; CANNOT be delegated; answerable when things go wrong
- **C — Consulted** — provides input BEFORE the decision; two-way communication; influences but doesn't veto
- **I — Informed** — notified AFTER the decision; one-way communication; not involved in making it
- **Exam move (mock Q1.2)** — Responsible = the DOER. The trap: many confuse with Accountable (the OWNER, signs off)

Why RACI Breaks at AI Scale · S17 · Air Canada chatbot ★★★★★

Three places RACI fails when AI is in the loop. 1. **R when AI acts** — AI generates outputs in real time, no human involved. RACI assumes R is a human who can be trained/replaced. R defaults to the team that built the system, but they made a design decision weeks ago, not a judgment about this specific output 2. **A when AI causes harm** — A must be a single role; AI cannot legally hold A. Air Canada tried to put A on the chatbot ("separate legal entity"); tribunal said no, pushed it back to the org. Most organizations haven't answered which internal role holds A 3. **Scale breaks C and I** — at 500 AI decisions/day, you can't Consult anyone on each. Customer-facing teams + legal cut out → problems accumulate undetected

- **Exam move** — Air Canada is the canonical case; if asked about RACI + AI, this is the story

Cross-block patterns — debates that write themselves

Innovation Theatre ↔ Real AI Capability (S5 ↔ S14)

- **Theatre** — symbols of innovation without structural change (S5)
- **Real capability** — change models actually implemented, decision rights actually moved (S14, S17)
- **Exam move** — diagnose when an "AI strategy" is mostly performative

Centralized vs Federated (recurring tension)

- **Org structure (S4)** — Horizontal/Centralized vs Vertical/Embedded teams
- **AI Labs (S5)** — Centralized vs Decentralized lab
- **Governance (S17)** — Centralized vs Federated vs Embedded
- **Exam move** — the SAME tension across three layers; show the pattern

Make/Buy/Partner ↔ Sustainability (S6 ↔ S2)

- **Hahn et al.** — temporal + spatial dependency
- **BUY** — fast economic gain, but vendor's social/env practices become yours
- **Exam move** — make sourcing decisions explicitly sustainability-relevant

Speed-vs-Rigor across the course

- **Org structure** — speed vs consistency
- **Change** — coercion (quick, resented) vs participation (slow, committed)
- **Governance** — federated (fast) vs centralized (rigorous)
- **PM** — agile (fast learning) vs waterfall (predictable delivery)
- **Exam move** — the answer is always *context-dependent*; name the trade-off explicitly

Human Oversight — the recurring spine

- **Floridi & Cowls** — explicability + autonomy
- **EU AI Act** — high-risk requires human oversight
- **Kahneman/cognitive surrender** — passive HITL fails
- **RACI breakdown at AI scale** — C and I cut out
- **Exam move** — oversight is a *design problem*, not a checkbox

Authors to cite (anchor list)

- **Hahn et al. (2014)** — Integrative sustainability framework (Core, S2)
- **Crawford (2021)** — AI ecosystem / planetary supply chain (Core, S2)
- **Gambacorta & Shreeti (2025)** — AI ecosystem & barriers (Core, S2)
- **Floridi & Cowls (2019)** — 5 AI ethics principles (Key, S3)
- **OECD (2019)** — AI Principles (Key, S3)
- **Bass & Riggio (2006)** — Transformational leadership (Key, S3)
- **Dierendonck & Nuijten (2011)** — Servant leadership (Key, S3)
- **Brown et al. (2005)** — Ethical leadership (Key, S3)
- **Edmondson (1999; 2020)** — Psychological safety, moral courage (Core, S10/S3)
- **Hannah & Avolio (2010)** — Moral courage (Supporting, S3)
- **Czakert & Wietrak (2025)** — New leadership for AI (Supporting, S3)
- **Wade & Tomoko (2024)** — Building AI ethics capability (Supporting, S3)
- **Blackman (2022)** — Operationalizing AI ethics (Supporting, S3)
- **Bouquet et al. (2026)** — AI innovation 4 paths (Key, S6/S7)
- **Kloppenborg & Lanning (2012)** — Portfolio management (Core, S7)
- **Bakker & Demerouti (2006/2007)** — Job Demands-Resources (Core, S10)
- **Karasek & Johnson (1986)** — Demand-control (Supporting, S10)
- **Deci & Ryan** — Self-Determination Theory (Core, S10)
- **Csikszentmihalyi** — Flow Theory (Key, S10)
- **Hackman & Oldham** — Job Characteristics Model (Core, S10)
- **Rousseau (1995)** — Psychological contract (Core, S10)
- **Ferrar & Green (2021)** — People Analytics definition (Core, S10/S11)
- **Kahneman (Nobel 2002) & Tversky** — System 1/2 (Core, S12)
- **Lewin** — 3-step change (Core, S14)
- **Kotter** — 8 steps + (with Schlesinger) 6 resistance approaches (Core, S14)
- **Hiatt** — ADKAR (Core, S14)
- **Stouten et al. (2018)** — Evidence-based 10 steps (Core, S14)
- **Ajzen (1991)** — Theory of planned behavior (Supporting, S14)
- **Pajuoja** — ManCo / GenAI adoption (Key, S13)
- **Gartner** — 85% AI projects fail stat (Key, S15)

Caveat

The colour code (🔴 Core / 🔵 Key / 🟣 Supporting) reflects weight in slides + the mock-exam questions. Importance shifts with the exam question — a Supporting framework can be the killer cite if the prompt happens to be about it. Don't let the colour decide what you actually write.